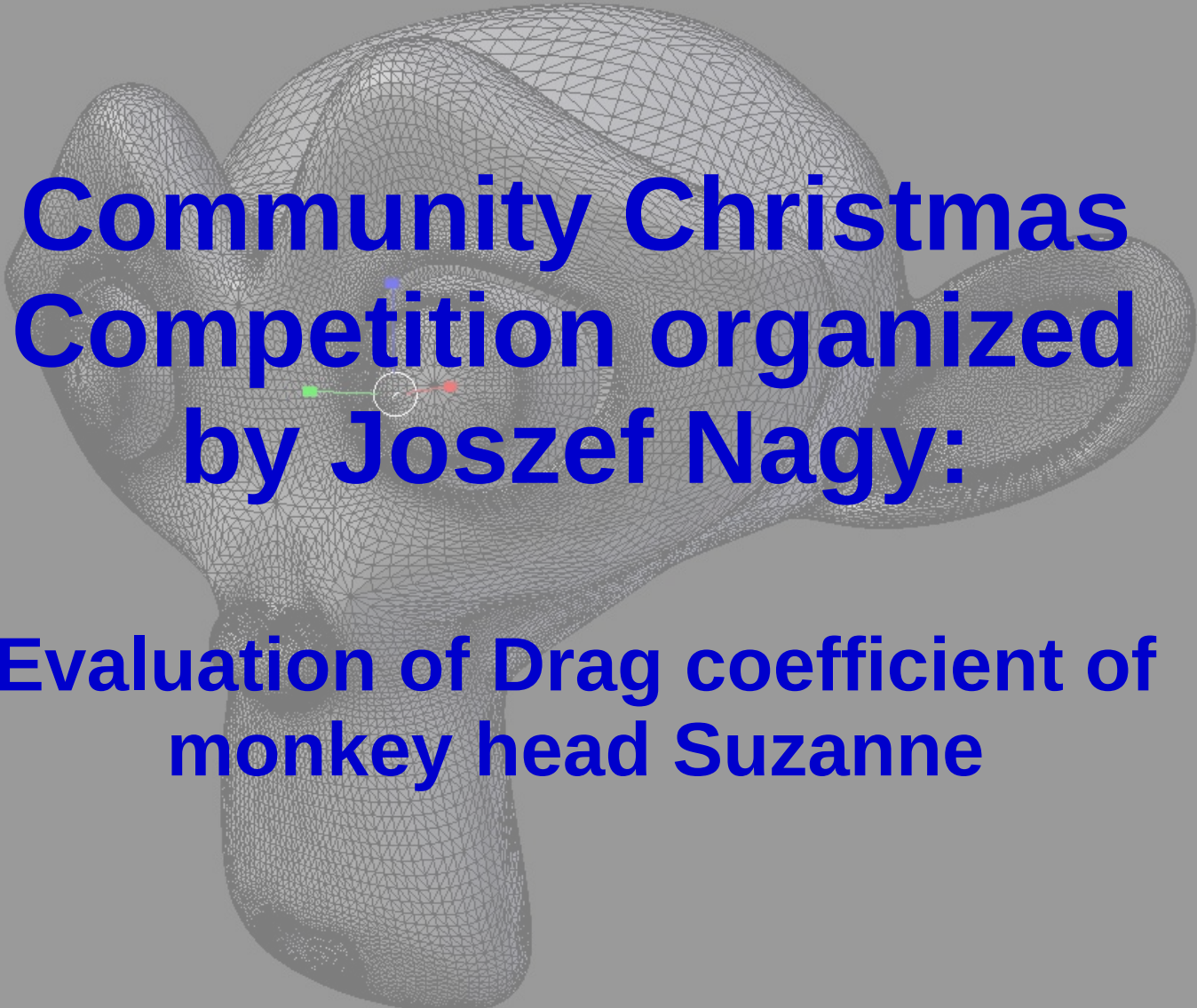




Community Christmas Competition organized by Jozsef Nagy:

Evaluation of Drag coefficient of monkey head Suzanne

Happy Foamer

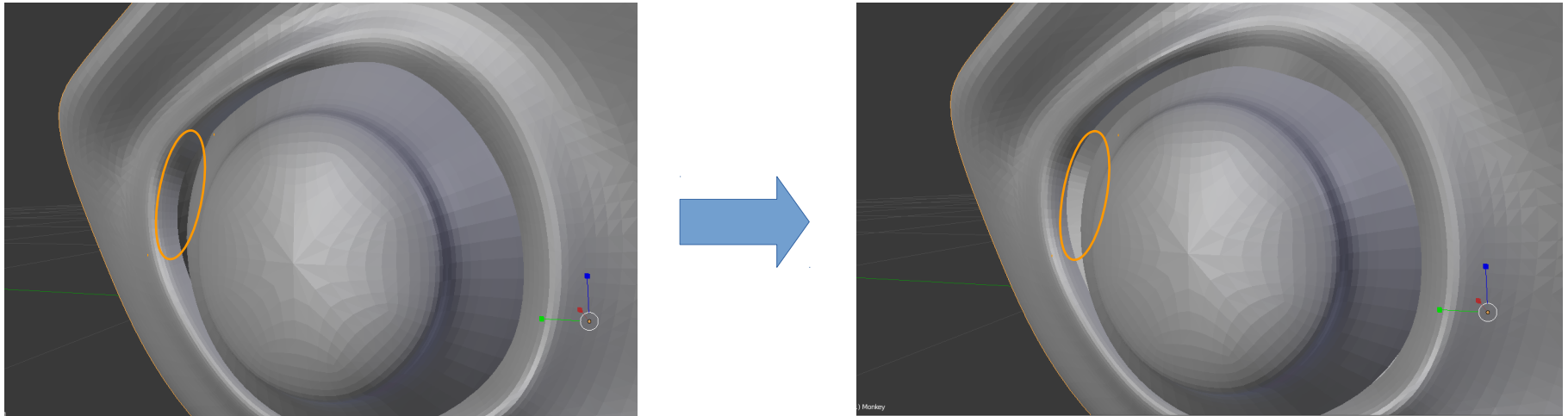
Community Christmas Competition with OpenFOAM® from Jozsef Nagy

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Preprocessing with OpenSource Tools

- Closing the gap between the eyes and the head with blender



- Derive model without intersecting parts by subtraction of monkey head from Cube (boolean operation) with blender → export stl-file
- Delete illegal vertices with meshLab → final stl-file for usage with SnappyHexMesh

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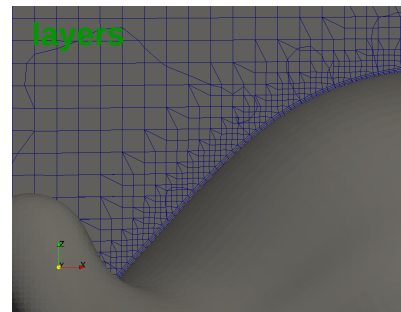
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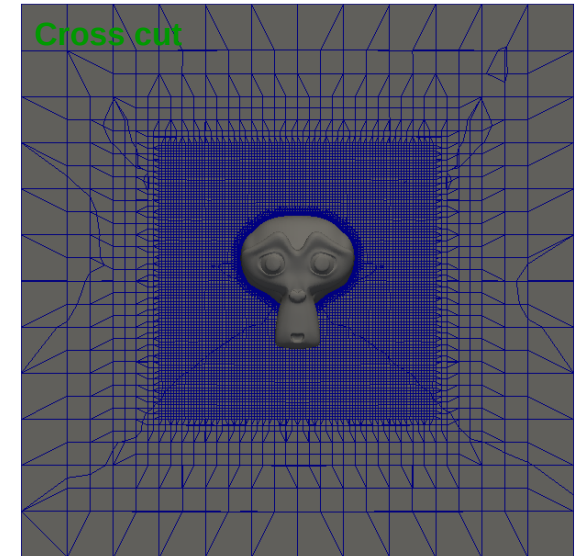
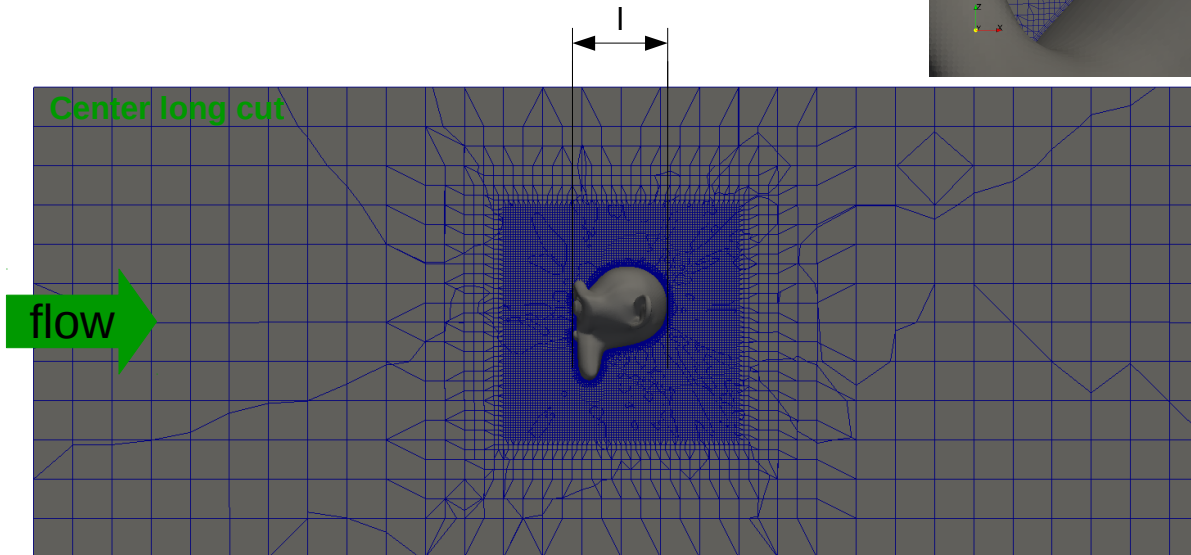
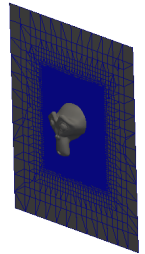
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Meshing with SnappyHexMesh

- Number of cells: 1.6 mio
- Snapshots of the mesh:



Cross cut
location



- Frontal area $A=2.553\text{m}^2$ and characteristic length $l= 1.6\text{m}$ for Re calculation was derived with help from paraview
- Solver: simpleFoam of OpenFOAM 5.0
- Turbulence model: kOmegaSST

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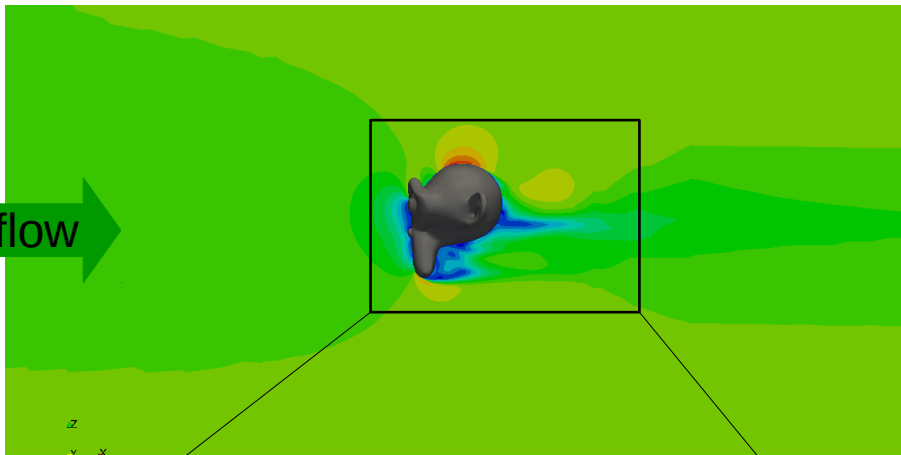
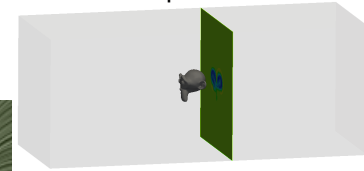
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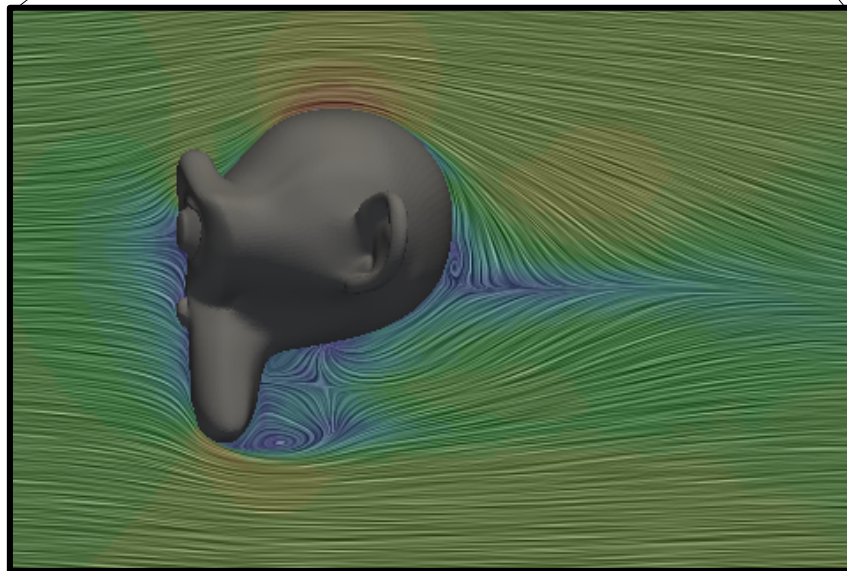
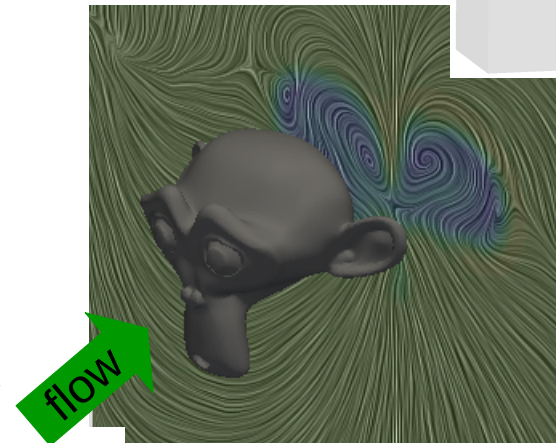
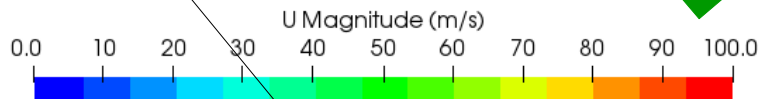
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Results for $Re=6.4 \cdot 10^6$

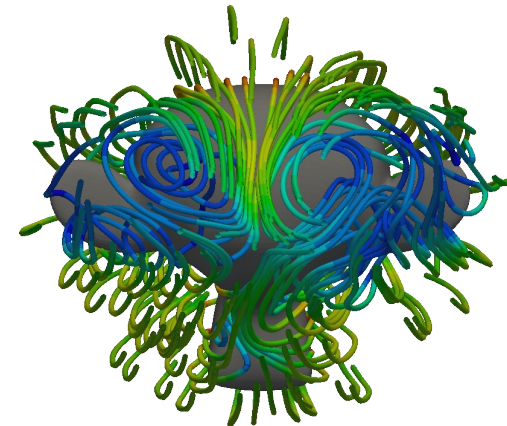
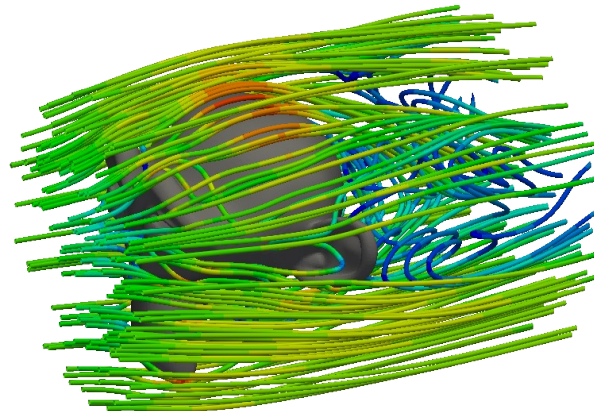
Location of plane



Central longitudinal cut



Vortex behind the monkey head
due to flow separation



View from behind

Happy Foamer

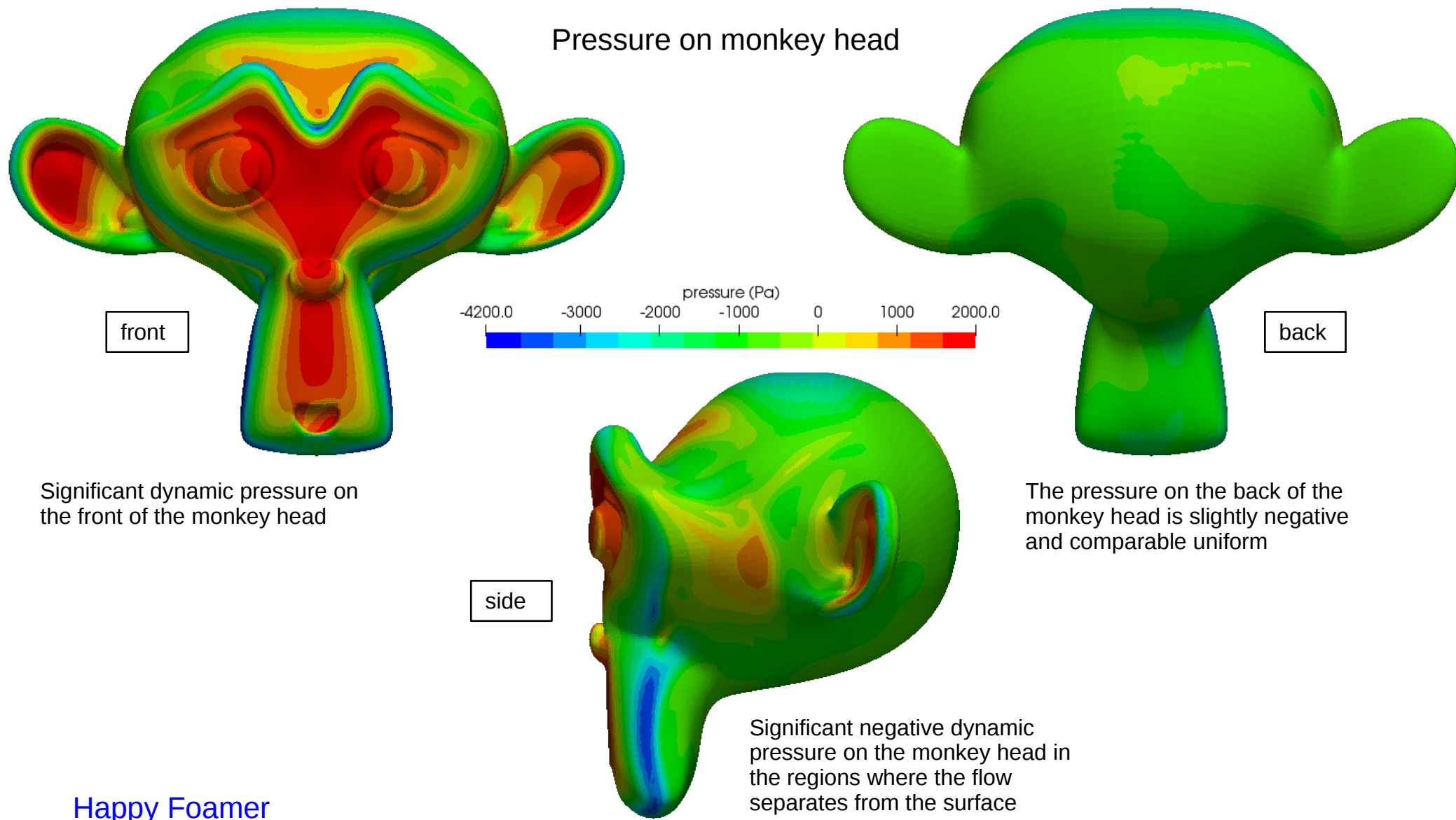
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Results for $Re=6.4 \times 10^6$



Happy Foamer

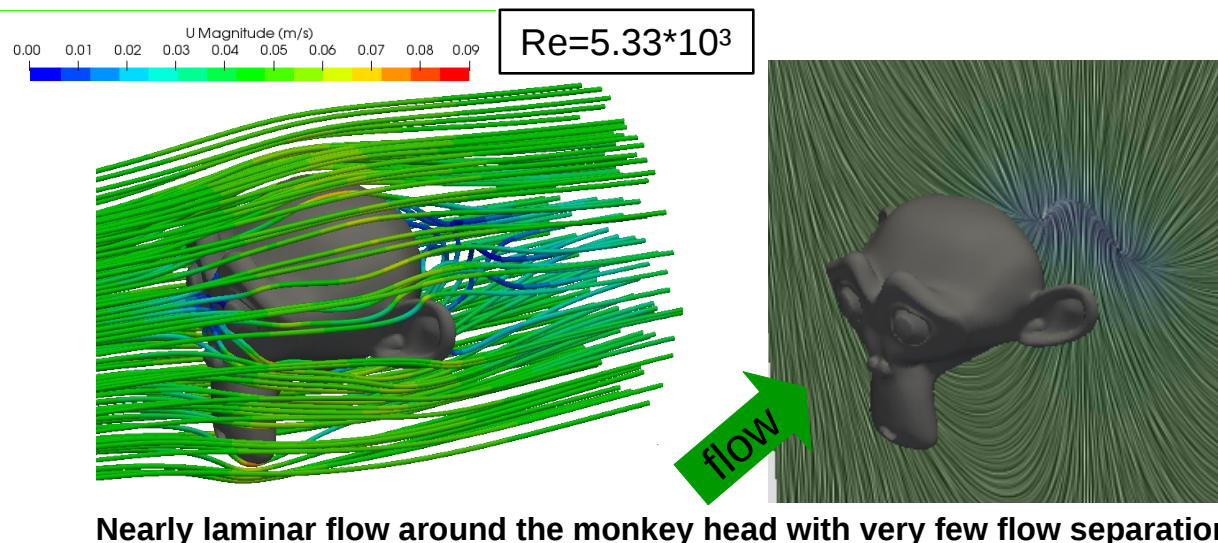
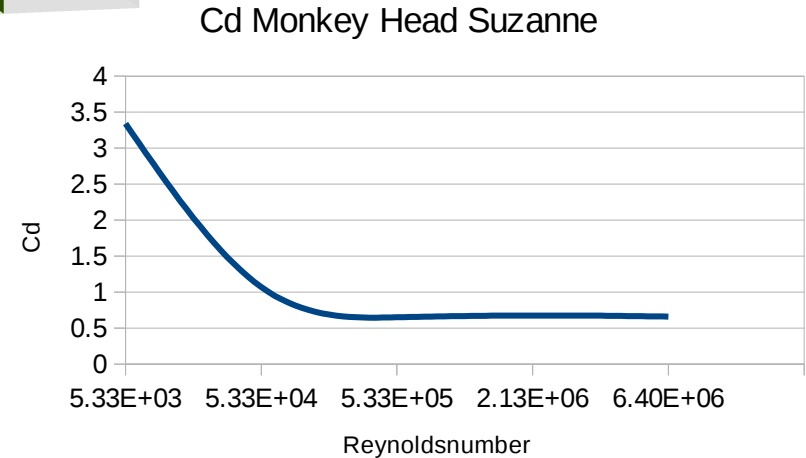
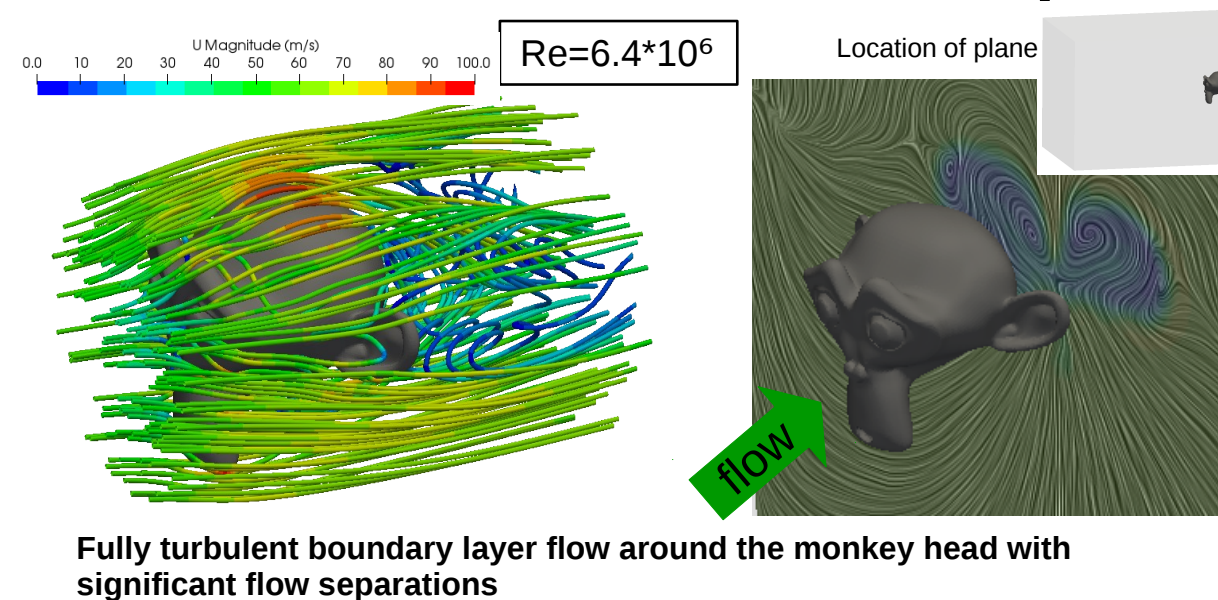
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Result: Cd depending on Re



With increasing Reynolds number the drag coefficient is decreasing till a fully turbulent boundary layer flow is reached. Once this is reached the drag coefficient is nearly constant for further increase of the Reynolds number. Anyhow one need to be aware of that the drag force itself is still increasing with growing Reynolds numbers.

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**Extra page with name
and contact address:**

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affiliation: none → private person
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